Ranked Differential Diagnosis Across Frontier LLMs: A Contamination-Controlled Neurology and Psychiatry Benchmark, and a Source-Grounded Retrieval Harness

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Abstract

We evaluate frontier large language models on open-ended clinical diagnosis from peer-reviewed neurology and psychiatry case reports, scoring the full ranked differential (top-1 through top-5) rather than a single label. We build the benchmark by using an LLM (DeepSeek V4 Pro) to turn published case reports into diagnostic challenges, then filter for answer leakage and for cases whose diagnosis is not derivable from the prompt when evaluated against the case study from which it was created; every retained case is a report published after every evaluated model's training cutoff, so it cannot have been seen in pretraining. On a 68-case hard set (cases DeepSeek V4 Flash fails closed-book), we evaluate seven frontier models—it plus DeepSeek V4 Pro, Gemini 3.5 Flash and 3.1 Pro, Claude Opus 4.8, and GPT-5.4 and 5.5—and report how they distribute correct answers across ranks. The central finding is that models differ not only in accuracy but in where they place the correct diagnosis: some surface it as their most confident (top-1) prediction, while others uncover it lower in a broader list, so the model ordering re-arranges as the rank cutoff grows. We then present ClinicalHarness, a retrieval-guided pipeline that pairs an LLM with PubMed/PMC retrieval and a cheap per-paper reader. A core feature is that it grounds each diagnosis in citable literature the clinician can verify—a cheap reader screens retrieved papers per case and returns a mean of about ten relevant sources (rejecting roughly 73% as irrelevant)—and it improves top-5 ranked accuracy for all three answerers we could run at pipeline scale, most where the base model is weakest.

1 Introduction

Medical LLMs perform well on board-style multiple-choice and short-vignette diagnosis (Singhal et al., 2023; Pal et al., 2022; Jin et al., 2019). Open-ended diagnosis from a real case report is harder: there are no options, the differential matters, and the answer is a specific named entity at a particular granularity. These systems should support a clinician rather than diagnose autonomously, and useful support is both thorough and verifiable: a short ranked list of plausible diagnoses helps more than a single best guess, and a diagnosis backed by sources the clinician can check helps more than an unsupported assertion. Most evaluations reward neither—they score a single label with no provenance. We therefore score the whole ranked differential (top-1 through top-5) and ask two questions. First, as a benchmark: how do frontier models compare when we credit a correct diagnosis appearing anywhere in their top-1 through top-5, on cases chosen to be hard and guaranteed post-cutoff? Second, as a systems question: can a retrieval harness raise a strong model's ranked recall?

The benchmark result is that frontier models re-order depending on the rank cutoff. A model that wins at top-1 may be overtaken at top-5 by a model that holds the correct entity lower in a broader differential. This is a statement about how a model distributes probability over a differential, not just about its competence, and it is invisible if one scores only a single label.

Contributions.

(1) A contamination-controlled, determinacy-audited benchmark of open-access neurology and psychiatry case reports, scored on the full ranked differential. (2) A cross-model evaluation of seven frontier models at top-1 through top-5 on a hard subset, with an interpretation of how their rank distributions differ, tempered by the fact that their training is undisclosed. (3) ClinicalHarness, a retrieval-guided pipeline that grounds each diagnosis in citable PubMed/PMC sources and improves top-5 ranked accuracy for every answerer we ran. Benchmark and code are released (§8).

Related work.

This work connects medical LLM evaluation (Singhal et al., 2023; Pal et al., 2022; Jin et al., 2019), retrieval-augmented generation (Lewis et al., 2020), self-consistency and multi-agent deliberation (Wang et al., 2023; Du et al., 2023), and LLM-as-judge evaluation (Liu et al., 2023; Li et al., 2024). Our angle is ranked-differential scoring of open-ended neurology/psychiatry diagnosis under contamination control.

2 Benchmark Construction

The benchmark turns published case reports into redacted diagnostic challenges, then audits each challenge for solvability. Figure 1 summarizes the pipeline.

Sourcing and licensing.

Cases derive from strictly CC-BY open-access neurology and psychiatry case reports indexed in PubMed Central; each source paper yields one challenge. Each case stores a structured gold diagnosis with aliases (for matching) and retains its source provenance (PMCID, DOI) so every challenge can be traced and re-audited.

Challenge creation.

We use an LLM (DeepSeek V4 Pro) to rewrite each source case report into a diagnostic challenge that withholds the diagnosis while keeping the clinical evidence needed to reach it. The model reads the full report and reproduces the presentation, workup, and findings up to the point of diagnosis, removing only the answer and any statement that gives it away.

Determinacy: is the answer in the prompt?

The key validity question is whether a challenge is determinate—whether every clinical fact required to reach the gold (a "discriminator") is present in the prompt. This can only be judged by reading the entire source case report against the challenge, since a discriminator that lives only in the source and is missing from the challenge makes the case unanswerable for any system, and so a benchmark defect rather than a model failure. The most common defect is asking for an entity whose defining test result is withheld. For example, a prompt might present a full phenotype, state that a gene panel "was sent," and ask for the specific gene—but never give the sequencing result, where the gene is not inferable from a phenotype shared with near neighbours (such as DJ-1/PARK7 versus PRKN versus PINK1 in autosomal-recessive early-onset Parkinson's). A genuine reasoning failure is the opposite: all discriminators are present and the model still picks the wrong entity. Only the second should count against a model.

Repair or drop.

DeepSeek V4 was used to validate determinacy: it reads the challenge, the gold, and the source full text and flags non-determinate cases. Each flagged case is then either repaired from the source—adding the missing deciding result (the sequencing result, antibody titre, or biopsy finding) so the gold becomes reachable, or reframing the question to the determinable next step—or dropped when no source-grounded repair is possible. Repairs never relax the gold on the evaluation set. Re-auditing one 349-case batch flagged about a quarter of cases; 34 were repaired from source and 16 were dropped.

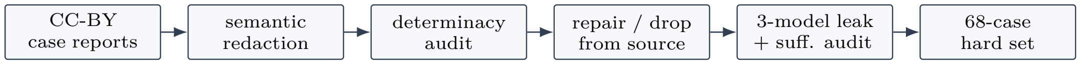

Figure 1: Benchmark construction. CC-BY case reports are turned into challenges by an LLM, audited for determinacy (is the answer derivable from the prompt?), repaired from the source or dropped, and—for the cross-model set—adversarially audited by three frontier models for answer leakage and insufficiency before a hand-checked drop/mend pass.

Adversarial audit of the cross-model set.

For the cross-model comparison we begin with cases that DeepSeek V4 Flash fails closed-book, then audit them with three frontier models—GPT-5.4, GPT-5.5, and Claude Opus 4.8—each scoring two independent defects: leakage (the prompt gives the answer away) and insufficiency (the prompt cannot determine the gold). Because automated auditors over-flag, every flag is reviewed by hand against the source under a fixed rule: drop a case only when the prompt directly reveals the target or the gold is not derivable; repair a case only when a source-grounded edit removes the leak or supplies the missing fact without changing the gold; otherwise keep it. Typical drops are cases where histology and molecular testing in the prompt already name the diagnosis; typical repairs delete a post-question sentence that reveals the outcome, or move a required result to before the question. The result is 13 drops and 11 source-grounded repairs, leaving the 68-case set used throughout.

Selection funnel and an economic-filter assumption.

The 68 cases are the survivors of a deliberate cost-saving filter (Figure 1). Across two post-cutoff batches we began with 399 case challenges and kept only the 81 that DeepSeek V4 Flash—a cheap, lower-capacity model—failed closed-book—discarding the 318 cases Flash solved—and the adversarial audit then reduced these to 68. The rationale is economic: running every frontier model on all 399 cases is costly, and the working assumption is that cases a cheap model already solves are unlikely to discriminate among stronger models, so concentrating the expensive frontier evaluation on the cheap model's failures saves budget while retaining the hard cases. This assumption is not guaranteed—a frontier model may fail a case Flash solves, or vice versa, so the filter could both miss discriminating cases and over-represent Flash-specific weaknesses. We flag it as a deliberate but unverified design choice; a fuller study would evaluate all models on the entire pool.

Why a ranked differential.

For the near-neighbour cases above, the correct entity is often already among a model's candidates even when its single best guess is wrong. Crediting a diagnosis that appears anywhere in a ranked top-5 measures whether the system surfaced the right answer at all, and avoids penalising it for the granularity of the single #1 choice. We score top-1 through top-5 throughout.

Composition.

Table 1 characterizes the set, and Figure 2 shows its etiologic-class frequencies. It is intentionally broad: cases span a broad range of conditions (Figure 2 shows the etiologic-class breakdown), and across the 64 source papers carrying author keywords there are 338 distinct keywords with almost no repetition. Patients span pediatric to geriatric (age median 48, range 4–95) and both sexes. Recent reports are not yet MeSH-indexed, so we report author keywords and derived classes in place of MeSH headings.

3 Cross-Model Benchmark Results

We score each model's bare (no retrieval) ranked differential at temperature 0 on all 68 cases, crediting a diagnosis that appears at rank $\leq n$. Scoring uses an alias-aware LLM judge with lexical fallback, using majority-of-3 judging because single-vote judging at temperature 0 is itself mildly stochastic.

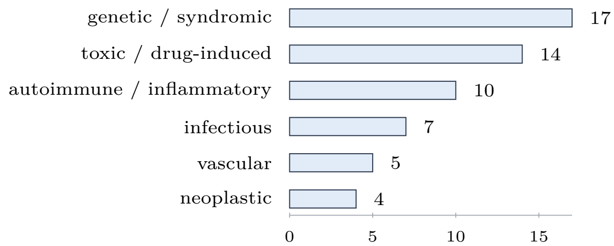

Figure 2: Etiologic-class frequency across the 68 cases (non-exclusive). Recent reports are not yet MeSH-indexed, so we use derived class labels and author keywords rather than MeSH headings.

Dimension	Distribution (of 68 cases)
Post-cutoff batches	two batches (33, 35); all published after every model's cutoff
License	CC-BY (68/68)
Patient sex	36 female, 28 male
Patient age	median 48, range 4–95 (pediatric to geriatric)
Author keywords	338 unique across 64 papers; long-tail (no concept recurs >2× except "case report")
Source venues	Cureus 24, Frontiers family 11, Elsevier 4, Hindawi 3, others

Table 1: Composition of the 68-case cross-model set: a long tail of specific conditions rather than a few concentrated topics.

3.1 Interpretation: models re-order as the rank cutoff grows

Model	top-1	top-2	top-3	top-4	top-5
GPT-5.4	47	53	55	56	57
Gemini 3.5 Flash	29	42	49	52	54
GPT-5.5	43	46	49	51	51
Claude Opus 4.8	33	41	46	48	50
DeepSeek V4 Pro	31	40	43	44	47
Gemini 3.1 Pro	29	38	41	41	44
DeepSeek V4 Flash	17	27	32	36	40

Table 2: Bare ranked-differential accuracy (count of 68) at each rank cutoff, temperature 0, majority-of-3 judge. The set is failure-selected (DeepSeek V4 Flash top-1 failures), so absolute numbers index difficulty, not general quality.

Table 2 is more revealing read across columns than down any one. We offer the following as interpretation, not mechanism.

Models differ in how peaked their differential is.

GPT-5.5 is nearly flat after top-3 (49 → 51 → 51): when it is right, it is right early, and extra ranks add little. Gemini 3.5 Flash is the opposite—it starts low at top-1 (29) but gains steadily to 54 at top-5, frequently having the correct entity but placing it lower in the list. The two cross over: GPT-5.5 leads Flash at top-1 (43 vs. 29) but trails it at top-5 (51 vs. 54).

The top-1 to top-2 jump separates the spread models.

The per-rank gains expose the mechanism. Gemini 3.5 Flash gains +13 from top-1 to top-2—the largest single-rank jump of any model—and +25 overall, climbing from sixth at top-1 (below DeepSeek Pro's 31) to second by top-3 and holding second at top-5, overtaking DeepSeek Pro, Opus, and GPT-5.5. A +13 jump means 13 of 68 cases where the model's rank-1 was wrong but its rank-2 was right: near misses in which the correct diagnosis was the model's close second, a ranking choice rather than ignorance. The two cheap "Flash" models climb most (Gemini Flash +25, DeepSeek Flash +23); the large GPT-5.x models climb least (GPT-5.4 +10, GPT-5.5 +8, the latter flat after top-3), behaving like committed single-answer models. A single-label leaderboard would place Gemini Flash fifth or sixth; on the differential it is second.

Further observations, offered without mechanism.

Two patterns stand out. Within a family the cheaper model can win: Gemini 3.5 Flash reaches top-5 54, ten points above Gemini 3.1 Pro (44), so capability tier and price do not predict differential quality here. And even the selection model recovers with ranks: DeepSeek V4 Flash defines the set—it fails these at top-1 by construction—yet recovers 40/68 at top-5, so for much of its "failures" the knowledge was in the differential and the failure was one of selection. We were genuinely surprised that the smallest, cheapest model (Gemini Flash) climbed from sixth at top-1 to second at top-5, and that the earlier GPT-5.4 edged the later GPT-5.5; but the training data and architectures of these models are undisclosed, so we can establish that their rank distributions differ without being able to say why. This is also a failure-selected, single-seed set, so fine rank distinctions should not be over-read.

4 ClinicalHarness

ClinicalHarness is a retrieval-guided diagnostic pipeline (Figure 3) that uses two distinct roles: a primary LLM that reasons about the case and produces the differential, and a cheap, separate reader LLM (DeepSeek V4 Flash in our runs) that screens individual retrieved papers. The split exists because most retrieved papers are irrelevant: the reader judges each paper against the current case and passes back only the small distillate it deems useful, so the primary LLM is never flooded with raw paper text and can read many papers per case on a small context.

1. Problem representation and initial differential (primary LLM).

The primary LLM reads the case and produces a structured problem representation and an initial, pre-retrieval differential of candidate diagnoses. A feature-indexed preset selector first maps the case to a diagnostic family (for example acute neurological emergency, infection/inflammation, or rare tumour), which conditions the prompt toward the relevant discriminators.

2. Query generation (primary LLM).

From the problem representation the primary LLM proposes targeted PubMed/PMC queries. These favour discriminator queries that separate the leading hypothesis from its closest mimics, rather than queries that merely confirm the leading hypothesis.

3. Retrieval in eval mode.

Queries run against PubMed/PMC. Eval mode enforces contamination control: the case's own source paper is excluded by PMCID, DOI, title, and generic-mimic filters, so the system cannot retrieve the answer. The harness may search for syndromes, discriminators, imaging patterns, or treatment guidelines, but never for the source itself.

4. Per-paper screening (reader LLM).

Each retrieved paper is screened in its own isolated reader call. The reader returns a compact structured note only when the paper is judged relevant to this case—a short relevant excerpt, the discriminators it supplies, which hypotheses it supports or refutes, any candidate diagnosis it raises that is not yet in the differential, and follow-up queries it suggests. The vast majority of retrieved papers contain no usable information for the case; those are discarded, and only the relevant distillates pass back. The full paper text never re-enters the primary LLM's context.

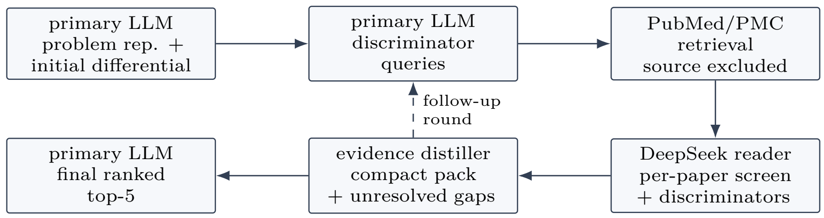

Figure 3: ClinicalHarness pipeline. The primary LLM forms an initial differential and targeted queries; retrieval runs in eval mode (the source paper is excluded); a cheap reader LLM screens each retrieved paper and returns only case-relevant findings; a distiller compiles the evidence and decides whether to run another retrieval round; the primary LLM then produces the final ranked differential.

5. Evidence distillation and adaptive rounds.

The relevant per-paper notes are compiled into a single evidence pack that summarises the decisive discriminators and the still-unresolved mimic pairs, and decides whether another retrieval round is warranted. When a discriminator remains unresolved, the system issues a follow-up query plan and runs another round (up to a bounded maximum); otherwise it finalises.

6. Final ranked differential (primary LLM).

The primary LLM receives the evidence pack together with its earlier problem representation and emits a ranked top-5 differential and a rank-1 final diagnosis. Retrieval may add literature-surfaced candidates and re-order the list relative to the pre-retrieval differential.

Development.

The harness components were developed and tuned on earlier validated case batches that predate this benchmark—roughly 350 development cases—using DeepSeek V4 Pro as a strong primary LLM to surface and fix failure modes. Tuning was frozen before the post-cutoff 68-case set was constructed, so no component was optimised on the evaluation cases.

5 Harness Results: Bare versus Harness (top-5)

We compare each model's bare top-5 (Table 2) against its top-5 inside the harness as the primary LLM, paired with the cheap DeepSeek reader and the same majority-of-3 judge, on the same 68 cases. Both sides are temperature 0.

Why not all models in the harness.

The harness is a multi-call pipeline—an initial differential, query generation, per-paper reader screens across several retrieval rounds, a distiller, and a final reranker—so it issues roughly an order of magnitude more model calls (and tokens) per case than a single bare query. At that volume, running Claude Opus, the GPT-5.x models, or Gemini 3.1 Pro as the primary LLM was cost-prohibitive; all remain affordable in the single-call bare setting. We therefore report all seven models bare (Table 2) and restrict the harness comparison to the Gemini Flash and DeepSeek answerers.

Do-no-harm fusion.

We combine the harness with the model's own differential conservatively: we keep the model's bare top-4 unchanged and let the harness fill only the fifth slot, with the single highest-confidence diagnosis the retrieval surfaced that the model did not already list. By construction this cannot demote a confident bare candidate—top-1 through top-4 are identical to bare—so the harness contributes exactly where it should, recovering a missed diagnosis into the last slot. This improves top-5 for all three answerers (Table 3), and the gain grows as the base model weakens—+1 for the strongest answerer (Gemini Flash), +2 for DeepSeek V4 Pro, and +3 for the weakest (DeepSeek V4 Flash)—consistent with retrieval helping

Model (primary LLM)		top-1	top-2	top-3	top-4	top-5
Gemini 3.5 Flash	bare	28	44	51	54	56
	fused	28	44	51	54	57
DeepSeek V4 Pro	bare	31	39	45	45	48
	fused	31	39	45	45	50
DeepSeek V4 Flash	bare	17	27	34	39	43
	fused	17	27	34	39	46

Table 3: Bare versus do-no-harm fused ranked accuracy (count of 68) at each cutoff, temperature 0, same majority-of-3 judge within each run. The fusion preserves the model's bare top-4 and fills only the fifth slot with the harness's single highest-confidence retrieval-surfaced diagnosis absent from the bare list, so top-1 through top-4 are identical to bare and only top-5 moves. Scores are within-run (the bare differential re-judged by the same judge), so they differ by ≤ 3 from the standalone bare scores in Table 2. The harness ran end-to-end only for the answerers shown; the others were evaluated bare only.

most where the base model has a genuine knowledge gap. (Substituting the harness's re-ordered differential wholesale instead of fusing it lowers ranked accuracy, by crowding the model's own correct diagnoses down the list; §6.)

Source-grounded diagnosis.

Beyond accuracy, a central feature of the harness is that every diagnosis arrives with the literature it rests on, so a clinician can check the reasoning rather than trust an opaque guess. The reader screens each retrieved paper against the case and returns only the relevant ones with the specific discriminators they supply: a mean of about 10 grounding sources per case (standard deviation ∼9.5, range 0–39), selected from a mean of ∼39 retrieved papers of which the reader rejects roughly 73% as irrelevant. These retained papers, attached to the final differential by PMCID/DOI, are the provenance a clinician can follow to verify each candidate.

6 Discussion

Two themes recur. First, differential structure is a model property: frontier models that look similar on a single label differ markedly in how they spread correct answers across a ranked list, and the appropriate metric depends on whether the consumer wants one label or a short differential. Concretely, the choice of metric re-orders the leaderboard—at top-1 the GPT-5.x models lead (GPT-5.4 first with 47, GPT-5.5 second with 43) while Gemini 3.5 Flash sits near the bottom (29); by top-5 Gemini Flash has climbed to second (54) and GPT-5.5 has fallen behind it (51)—so a single-label evaluation mis-ranks exactly the models that are most useful when the deliverable is a differential for a clinician to weigh.

Second, augmentation must be integrated conservatively. Replacing a strong model's differential with retrieval output can crowd its own correct diagnoses down the list, because a model with rich internal medical knowledge already holds a broader differential than a few distilled passages convey. A do-no-harm fusion that preserves the model's confident candidates and lets retrieval add only a diagnosis the model missed avoids this and yields a consistent top-5 gain (Table 3), largest where the base model is weakest—supporting the reading that external evidence complements parametric knowledge at the margin rather than replacing it. How far that margin extends—how much a stronger reader, broader retrieval, or a multi-slot fusion can add without reintroducing interference—is open work.

7 Limitations

The hard set is failure-selected, so absolute scores index difficulty rather than general quality, and small rank gaps are within judge noise. Mechanistic claims about why model differentials differ are out of reach because training details are undisclosed. Published case reports are retrospective and selected for novelty, so this is a test of diagnostic reasoning from curated evidence, not a clinical decision-support trial. The harness fusion is deliberately conservative (it touches only the fifth slot); a multi-slot fusion could add more but risks reintroducing the interference that single-list substitution showed.

8 Data, Code, and Ethics

The benchmark and ClinicalHarness code are released at https://github.com/Santosh-Gupta/ClinicalOrchestra. Runs persist prompts, model responses, retrieval provenance, and scoring records so every reported number is traceable to an artifact. This work evaluates retrospective, published case reports; it is not a clinical decision-support system and must not be used for patient care without prospective validation and governance.

References

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. arXiv preprint arXiv:2305.14325, 2023. URL https://arxiv.org/abs/2305.14325.

Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing, 2019. URL https://aclanthology.org/D19-1259/.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In Advances in Neural Information Processing Systems, 2020. URL https://arxiv.org/abs/2005.11401.

Tianle Li, Wei-Lin Chiang, Evan Frick, Lisa Dunlap, Tianhao Wu, Banghua Zhu, Joseph E. Gonzalez, and Ion Stoica. Justice or prejudice? quantifying biases in LLM-as-a-judge. arXiv preprint arXiv:2410.02736, 2024. URL https://arxiv.org/abs/2410.02736.

Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-Eval: NLG evaluation using GPT-4 with better human alignment. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023. URL https://aclanthology.org/2023.emnlp-main.153/.

Ankit Pal, Logesh Kumar Umapathi, and Malaikannan Sankarasubbu. MedMCQA: A large-scale multi-subject multi-choice dataset for medical domain question answering. In Proceedings of the Conference on Health, Inference, and Learning, volume 174 of Proceedings of Machine Learning Research, pp. 248–260. PMLR, 2022. URL https://proceedings.mlr.press/v174/pal22a.html.

Karan Singhal, Shekoofeh Azizi, Tao Tu, S. Sara Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Tanwani, Heather Cole-Lewis, Stephen Pfohl, Perry Payne, Martin Seneviratne, Paul Gamble, Chris Kelly, Abdelrahman Babiker, Nathanael Schärli, Aakanksha Chowdhery, Philip Mansfield, Dina Demner-Fushman, Blaise Agüera y Arcas, Dale Webster, Greg S. Corrado, Yossi Matias, Katherine Chou, Juraj Gottweis, Nenad Tomasev, Yun Liu, Alvin Rajkomar, Joelle Barral, Christopher Semturs, Alan Karthikesalingam, and Vivek Natarajan. Large language models encode clinical knowledge. Nature, 2023. doi: 10.1038/s41586-023-06291-2. URL https://www.nature.com/articles/s41586-023-06291-2.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In International Conference on Learning Representations, 2023. URL https://arxiv.org/abs/2203.11171.

A Benchmark construction pipeline components (Figure 1)

The six stages of Figure 1, end to end:

CC-BY case reports. The raw input: strictly CC-BY open-access neurology and psychiatry case reports indexed in PubMed Central, each retained with its full text, PMCID, and DOI. One source paper becomes one challenge.

Semantic redaction. An LLM (DeepSeek V4 Pro) rewrites each report into a diagnostic challenge that reproduces the presentation, workup, and findings up to the point of diagnosis while removing the stated diagnosis and any sentence that gives it away (final "the diagnosis was X" lines, disease-naming treatment, and similar tells).

Determinacy audit. A check, made against the full source, that every clinical fact required to reach the gold (every "discriminator") survives in the redacted challenge—so the case is solvable from the prompt alone and a wrong answer reflects reasoning rather than a fact missing from the prompt.

Repair / drop from source. Non-determinate challenges are either repaired from the source—re-inserting the missing deciding result (a sequencing result, antibody titre, or biopsy finding) or reframing to the determinable next step—or dropped when no source-grounded repair is possible. Repairs never relax the gold.

3-model leak + sufficiency audit. For the cross-model set, three frontier models (GPT-5.4, GPT-5.5, Claude Opus 4.8) independently flag two defects per case—leakage (the prompt reveals the answer) and insufficiency (the prompt cannot determine the gold)—and every flag is reviewed by hand against the source before any drop or repair.

68-case hard set. The surviving challenges, restricted to cases the cheap DeepSeek V4 Flash fails closed-book, form the 68-case evaluation set used throughout. The harness pipeline (Figure 3) is described stage-by-stage in §4.